On the Automated Classification of Tidal Features in Galaxy Images

Daniel C. Bruwel★

May 2023

ABSTRACT

This paper presents a study of the automated classification of tidal features using convolutional neural networks (CNNs). Understanding tidal features gives insight into the merger history of galaxies, which is a crucial mechanism in the hierarchical model for galaxy formation. Manual classification of tidal features is feasible for small scale samples but becomes increasingly more difficult for large scale astronomical surveys. The primary aim of this paper is to study the effects of image processing on the learning rate and accuracy of CNNs. This study uses data from the $r-$band of the Hyper Suprime-Cam Subaru Strategic Program (HSC-SSP) survey and examines the performance of the AlexNet architecture on linearly scaled, log scaled, and highly preprocessed images. We find that the network using highly preprocessed images learn faster but reach a lower maximum accuracy, and that using image processing techniques that retain all the original information do not necessarily reach the same final accuracy. We find that the best image processing method was log scaling, with a Area Under the Curve of the Receiver Operating Characteristic (AUC-ROC) of 0.69. Additionally, we find that many of the classification errors results from spiral type galaxies, and image contamination (e.g. stars and image artifacts).

Key words: methods: data analysis – galaxies: evolution – galaxies: interactions – galaxies: structure – galaxies: statistics

Note: This manuscript is not published or endorsed by MNRAS. It is formatted using their LaTeX template for convenience.

1 INTRODUCTION

Modern theories of cosmology rely on the hierarchical structure formation theory. This theory assume that the Universe initially formed with small-scale Gaussian density fluctuations which were amplified during the inflation period. These density fluctuations grew and merged to form dark matter halos, which proceed to accrete gas to form dwarf galaxies. Lager galaxies such as ellipticals and spirals formed when these dwarf galaxies merged (Cole et al. 2000). This theory is supported both by direct simulations (e.g. Blanton et al. 2000) and semi-analytic models (e.g. Cole et al. 1994, 2000). Additionally, there is strong evidence from Steinmetz & Navarro (2002) that galaxy morphology is closely related to gas accretion and galaxy merger history, finding that: (i) disk and bulge systems, which are galaxies that have dense collections of starts in their centres and flat, circular regions of stars around them, arise from the smooth accretion of gas onto the remnants of major mergers (ii) Lyman break galaxies, which are high Redshift (and therefore very old) galaxies, include ongoing merging events associated with the formation of spheroid galaxies (iii) field ellipticals, which are isolated featureless galaxies, result from dissipative disk mergers.

Tidal features can be broadly classified as streams and shells, which can be seen in Figure 1. It has been established (e.g. Pop et al. 2018; Toomre & Toomre 1972; Tal et al. 2009) that tidal features are closely related to merger events. Tal et al. (2009) suggests that shells can form from the accretion of small galaxies by massive galaxies. However, Pop et al. (2018) suggests that these shells may arise from major merger events. There is general agreement (e.g. Tal et al. 2009; Hendel & Johnston 2015) that streams are formed from the merging of small, high angular momentum, infalling galaxies. As a result, being able to classify tidal features is vital in the study of merging events. Additionally, detailed surveys of tidal features will allow for been much success using machine learning to automatically classify galaxy morphology (e.g. Cheng et al. 2020; Walmsley et al. 2019), with Cheng et al. (2020) finding that convolutional neural networks have the highest success rate. Walmsley et al. (2018) and Sánchez et al. (2023) have had some success using CNNs to classify tidal features. However, Walmsley et al. (2018) was trained on data that is heavily biased towards bright systems, and Sánchez et al. (2023) was trained on simulated data and had a much lower success rate when applied to actual data and illuminates the need to build a network trained on real data.

As outlined by Sánchez (2021), automatic detection of tidal features is very challenging for a number of reasons (i) there are a low number of identified tidal features, leading to a small, unbalanced, training sample that requires techniques such as data augmentation Taylor & Nitschke (2018) and oversampling Buda et al. (2018), (ii) tidal features are subjective and the labels used are often subjective, (iii) tidal features have a low surface brightness and are often not easily visible against noisy backgrounds.

The primary aim of this project was to use real data with different image processing techniques to train multiple CNNs to classify tidal features, and to test if the accuracy and learning rate of these CNNs depended on the image processing.

It was hypothesised that the learning rate would depend on image processing, with more effectively processed images having a greater learning rates, and that the asymptotic accuracy would depend on the amount of information retained by the image processing, with methods that loose no information performing optimally.

Section 2 discusses the data used in the project. Section 3 discusses the different image processing methods used for the networks. Section 4 discusses the neural network architecture and hyperparameters. The results are analysed in section 5. Finally, section 6 has

© 2023 The Authors

a brief discussion on performance and improvements that could be made to the CNNs.

2 SAMPLE DATA

This project used wide field data from the $r-$band of the Hyper Suprime-Cam Subaru Strategic Program (HSC-SSP). The $r-$band was selected as Yoon & Lim (2020) suggested that it was the most optimal band to use for tidal feature detection. Each image was 512×512 pixels corresponding to a sky area of $\approx 86 \times 86 \text{arcsec}^2$ (conversion from Aihara et al. (2017)). The images are all processed by the HSC-SSP "pipeline" which (amongst other things) performers background subtraction and contamination detection (Bosch et al. 2018).

A sample of this data containing 434 images was manually classified by (Rutherford et. al in prep) (examples in Figures 1,3,9,10,11) who detected 134 features ($\approx 31\%$) these were further classified into streams (79 identified) and shells (72 identified). These images where manually rechecked and all highly corrupted images, or images that did not contain the entire galaxy were removed. In total there were 18 of these. This manual removal is valid as it is automatable.

In its current form there is not enough data to successfully train the CNNs. Additionally, there is a large class imbalance which can cause the neural network to classify all samples as featureless (Buda et al. 2018). To solve this, all of the feature data was flipped and rotated to multiply the sample size by 8. This yields 1072 feature images. The featureless data was only rotated, multiplying by sample size by 4, yielding 1128 images. As suggested by Buda et al. (2018), the classes should have the same size. So, random images where removed from each dataset to balance the classes, with care taken to not fully remove any images (i.e. there is always at least one version of any image in the dataset), yielding a final dataset of 1000 feature images and 1000 featureless images. The data was then split into a training and validation set with a training:testing ration of 80 : 20, where considerations were made to ensure that the same galaxy (with a different augmentation) didn't appear in both the training and validation set.

3 IMAGE PROCESSING

3.1 Scaling

It is common practice in astronomy to apply a function to the brightness values of each pixel to scale the images and make low brightness features appear more obvious. A variety of different scaling methods and parameters where tested. Ultimately linear and logarithmic ($a = 1000$) scaling were selected, an example of this can be seen in Figure 1. Linear scaling was chosen as it does not augment the brightness values in any way and is a more "true" representation of the information. Logarithmic scaling was used because it makes the features obvious while still being an invertible process (so retains all the information). The formula used for logarithmic scaling was

$$\frac{\log(1000x + 1)}{\log(1001)} \tag{1}$$

3.2 Preprocessing

A preprocessing method was developed to remove unnecessary background information, such as atmospheric noise and other stars, and

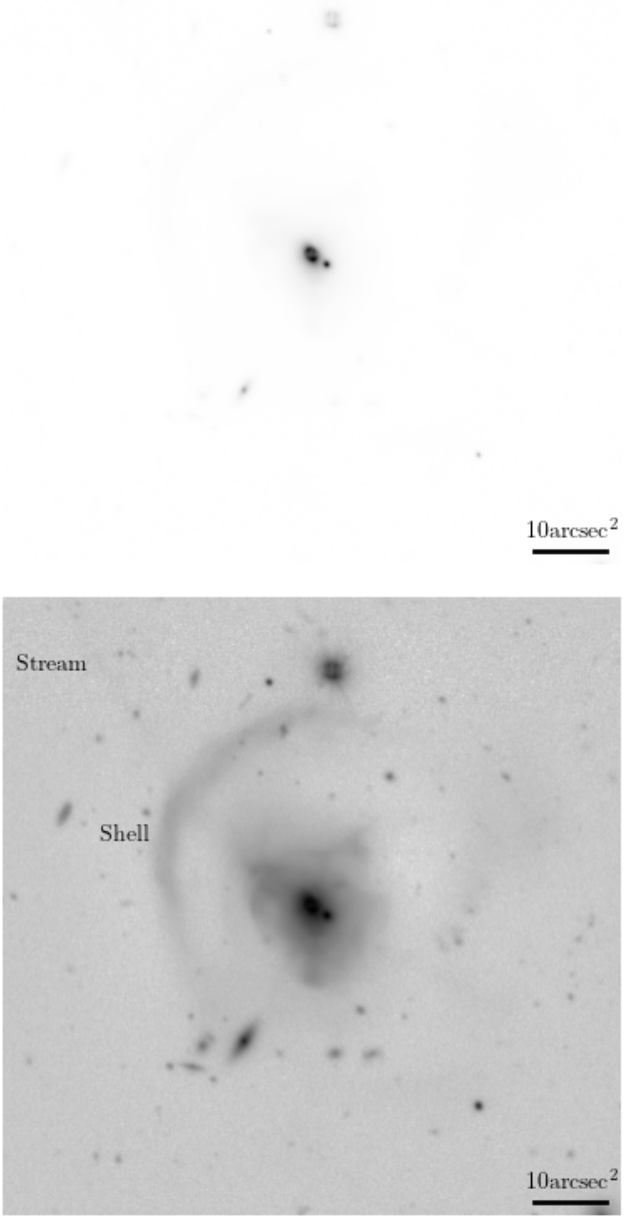


Figure 1. Inverted images of a galaxy (id = 55245) with shell and stream (top) linearly scaled (bottom) logarithmically scaled.

exaggerate features. To do this, a raw image was taken, the median brightness was calculated as a good approximation of the background brightness. The robust standard deviation was calculated as IQR/1.35, where IQR is the interquartile range, this was done as it is less affected by the bright pixels in the centre of the galaxy. For each pixel, the number of robust standard deviations above the median was calculated. This standard deviation data was logarithmically scaled to exaggerate features. To create a mask, a threshold of 0.5 was applied to the scaled image, this value was chosen heuristically based on some sample images with obvious features and could potentially be lowered to retain more information. The area of each connected region was calculated, any region that had an area less than 140arcsec^2

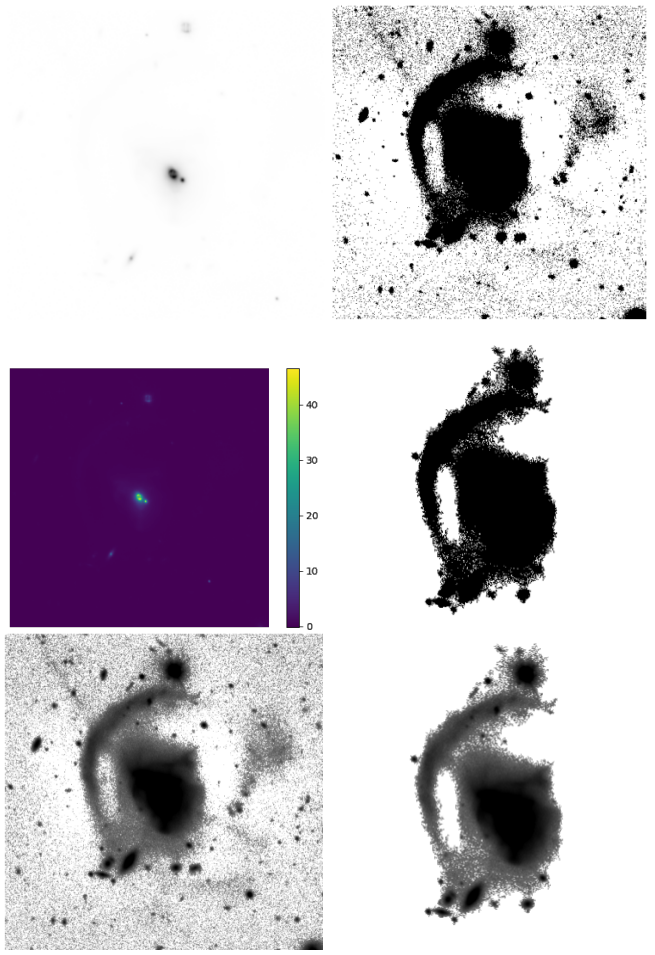


Figure 2. (Top left) Linearly scaled (raw) image (centre left) Heat map of number of robust standard deviations from median (bottom left) Log scaled number of standard deviations (top right) Threshold mask of 0.5 (centre right) Image with all small regions of connected pixels removed (bottom right) centre right used as a mask on bottom left

was removed. This mask was then applied to the scaled image to get the final preprocessed image. This process can be seen in Figure 3.2.

A few feature images where examined. It is noted that this process is not reversible, and information is lost. As seen in Figure 3, for sufficiently low surface brightness features the threshold can fully remove the tidal feature. Lowering the threshold or applying smoothing is possible, but the final image would retain more background noise.

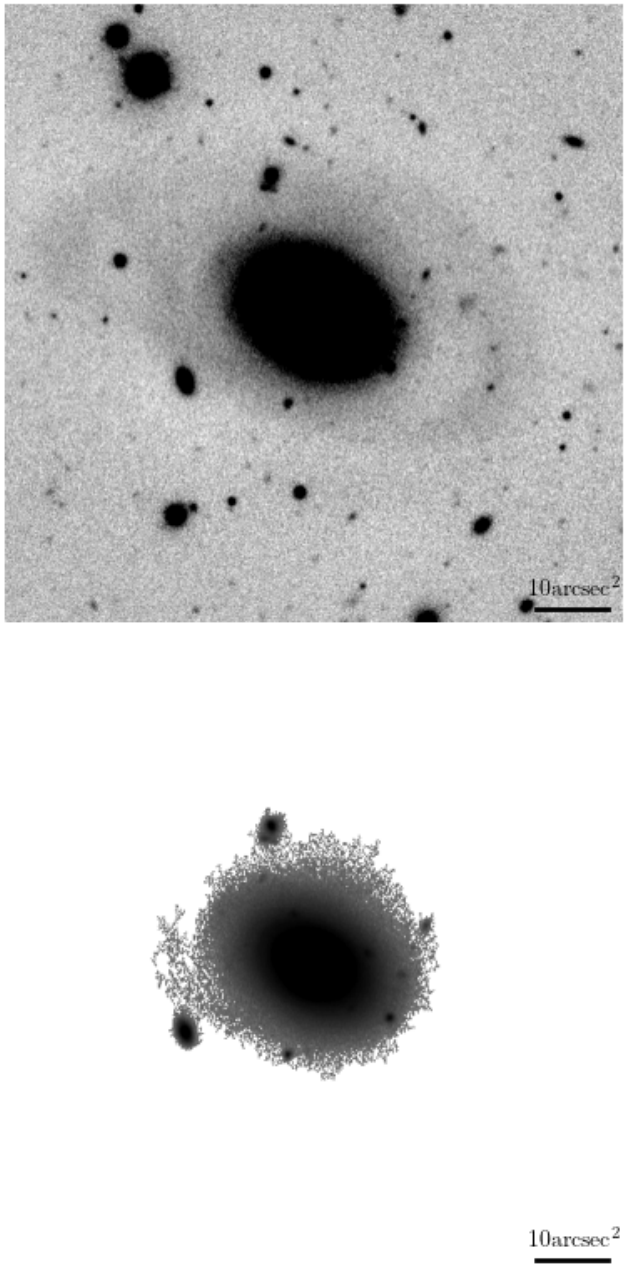


Figure 3. (top) Z-scaled image of a galaxy (id = 511892) with obvious shell feature (bottom) same galaxy after prepossessing is applied, shell is removed.

4 THE CONVOLUTIONAL NEURAL NETWORK

The CNN was built in python using the tensor-flow packager Keras (Abadi et al. 2015). This was chosen as it allows for quick modification of the CNNs hyperparameters and architectures.

The architecture chosen for the CNNs was based off AlexNet Krizhevsky et al. (2017). The architecture for the CNNs can be seen in Figure 4. The CNNs have 5 convolution layers with `relu` activation functions, which are zero below zero and linear above zero, and decreasing kernel sizes. These initial large kernels are good for detecting large scale features, where the final small kernels are good for detecting smaller scale features. Additionally there are max-pooling layers after the first two convolution layers which help decrease the size of the image. After the final convolution layer, the image is flattened and passed through 2 fully connected layers with `relu` activation functions and decreasing size until one final node with a `sigmoid` activation function. The final network has 1, 606, 941 trainable parameters.

It is important to note that the CNN architecture used is not optimized for this particular task.

A number of different optimizers (`adam` and `Least Squares (experimental)`) and learning rates (10^{-4}, 10^{-5} and 10^{-6}) where tested. Ultimately the CNN used the `adam` optimizer with a learning rate of 10^{-5}. The model was complied with a binary cross entropy loss function, which quantifies the

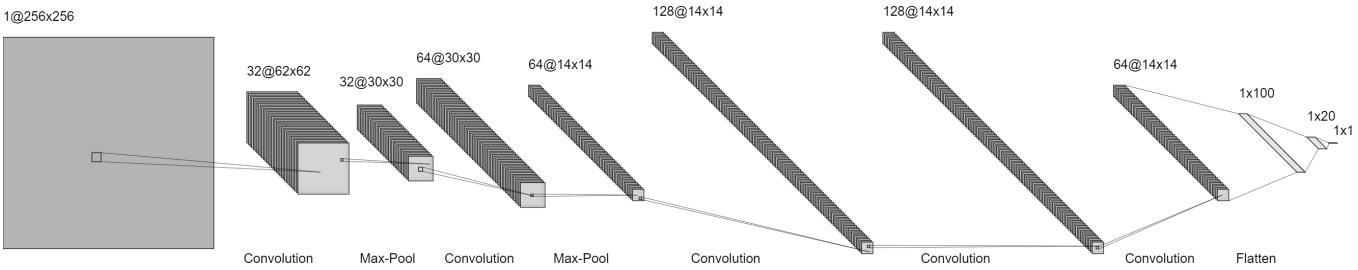


Figure 4. The architecture used for the CNNs.

difference in predicted and actual values, and the accuracy and area under ROC curve (AUC-ROC) were used as additional metrics of performance.

Early stopping with a patience of 3 was implemented into the CNN monitoring loss, which causes the network to terminate training when the loss of the network had not improved for 3 epochs. Additionally, checkpointing was implemented to save the best performing epoch. All networks where initialised to run until early stopping was triggered (loss stopped decreasing).

5 RESULTS

5.1 Performance of the different networks over time

We first examine the performance of the different networks as a function of the number of epochs that they have run through. In Figure 5 we see that all classifiers started with a loss of $0.69 \approx \ln(2)$ which was expected for a randomly initialised classifier with equal class sizes. The preprocessed images initially out performed the log scaled images which outperformed the linearly scaled images. However, by 15 epochs, the preprocessed images stopped due to the early stopping. By $\approx$ 17 epochs the log scaled classifier had outperformed the initially better preprocessed classifier. We observe that the unscaled classifier performs the worst and early stops at 22 epochs, indicating that it had learned all it could from the data. It is possible that with more data the classifier would have continued to improve.

We observe similar trends in Figure 6 when examining the accuracy and the AUC-ROC where the linearly scaled images perform worse than the preprocessed and log scaled images. It is interesting to note that the initial superiority of the preprocessed images is unclear under these metrics.

We now examine the performance of the best classifier from each preprocessing method. Figure 7 shows the ROC curve for each preprocessing method, we again see that the log scaled images performed the best, followed by the preprocessed images, and finally the linearly scaled images. We observe that despite the significantly superior performance of log scaling, it only had a ROC-AUC of 0.69 which is just below the standard acceptable score of 0.7 Mandrekar (2010).

5.2 Performance of the Log Scaled classifier

The overall metrics of the best log classifier are summarised in Table 1

5.2.1 *Generalizability*

We examine the confusion matrices of the classifier on both the training and testing data. Note that values have been standardized to

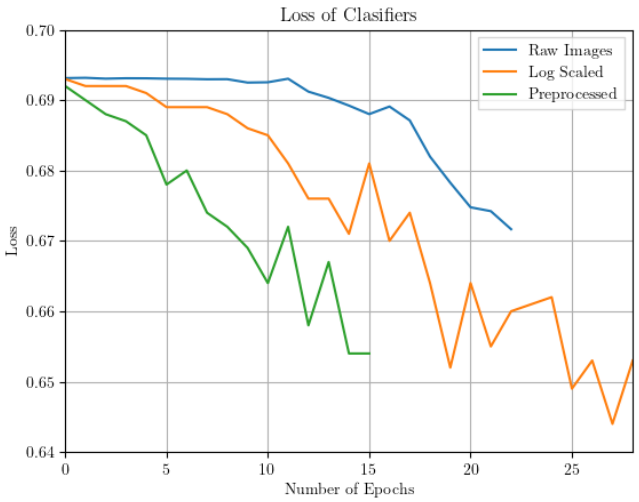


Figure 5. The loss (binary cross entropy) of each classifier as a function of number of epochs.

Table 1. Performance of the CNN with log scaling.

Accuracy	68%
AUC-ROC	0.68
Positive predictive value	64%
True positive rate	83%
F_1	72%

estimated performance on a dataset that has 75 data points, 25 with features (i.e. the sample distribution for the validation data). This is because confusions matrices are dependent on the feature ratio. We see in Figure 8 that the confusion matrices were very similar, indicating that the network generalised well. We notice a slight decrease in the true positive rate, and a slight increase in the true negative rate that results from $\approx$ 2 additional image misclassifications.

5.2.2 *Feature specific performance*

Due to the high generalizability of the network, we examine the performance of the network on all feature examples (both those in the training and the testing set). Of the 79 streams, 59 were classified successfully (TPR = 75%). Of the 72 shells, 55 were classified successfully (TPR = 76%). We can see that the network had no strong bias to classifying streams or shells, contrary to Sánchez et al. (2023).

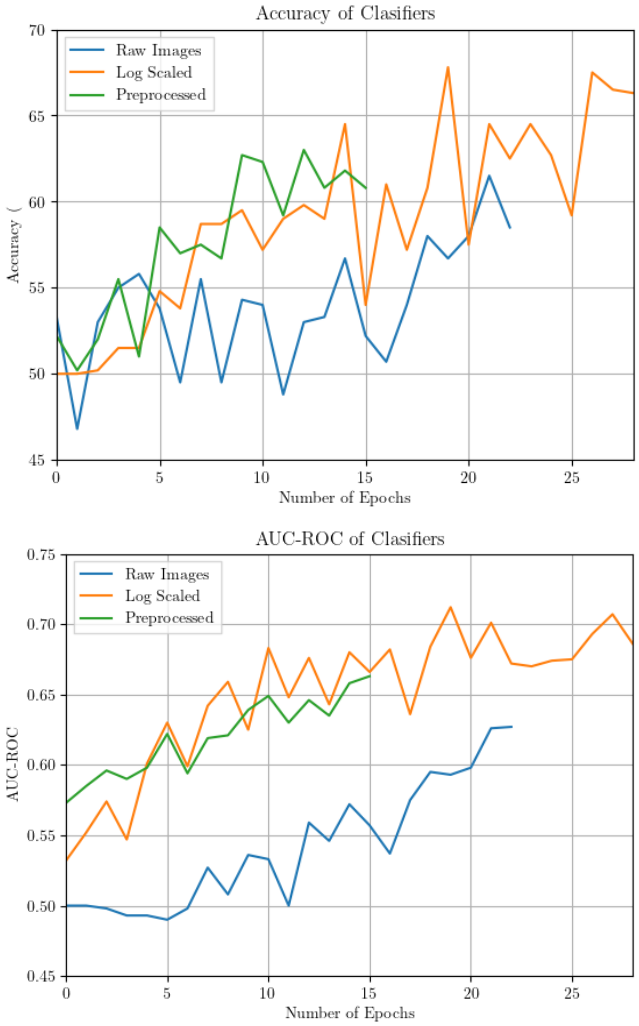


Figure 6. (top) accuracy of each classifier as a function of epochs (bottom) AUC-ROC for each classifier as a function of epochs.

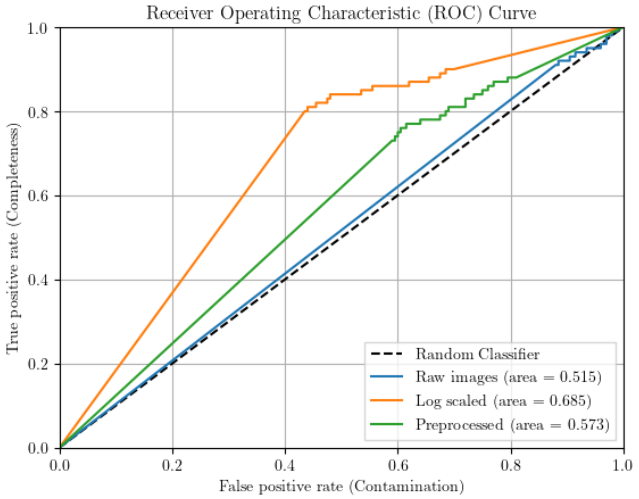


Figure 7. The ROC curve for each image processing method.

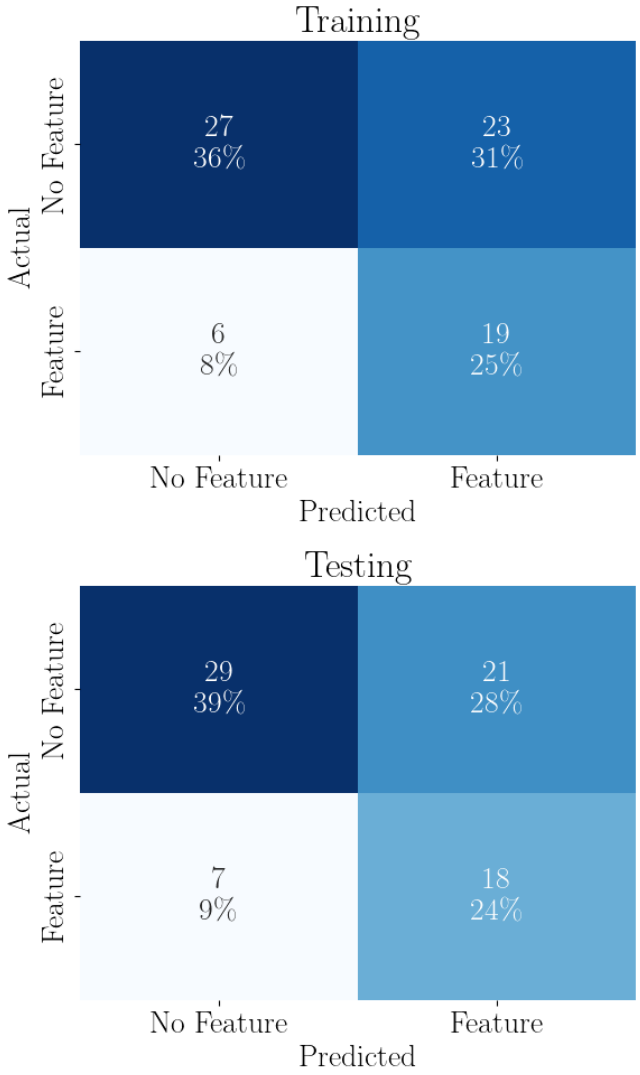


Figure 8. (top) confusion matrix of the training data (bottom) confusion matrix of the testing data.

5.2.3 Examples of false negatives

We manually examined the false positives. We observed a range of misclassifications, Figure 9 shows an example of a "reasonable" misclassification where the tidal feature is difficult to see in the log scaled image and the CNN misclassified it as featureless. Conversely, Figure 10 shows a "unreasonable" misclassification where the tidal feature is obvious but the CNN classified it has featureless. Heuristically it would appear that the network is bias to detecting obvious features, but is even inaccurate with these.

5.2.4 Examples of false positives

Figure 11 shows some examples false positives. One of the misclassifications has spiral arms, and the other has a camera artifact.

A manual check shows that of the 109 false positives in the training data, 10 had spiral arms, and 13 had artifacts. And of the 21 false positives in the testing data, 2 had spirals and 4 had artifacts.

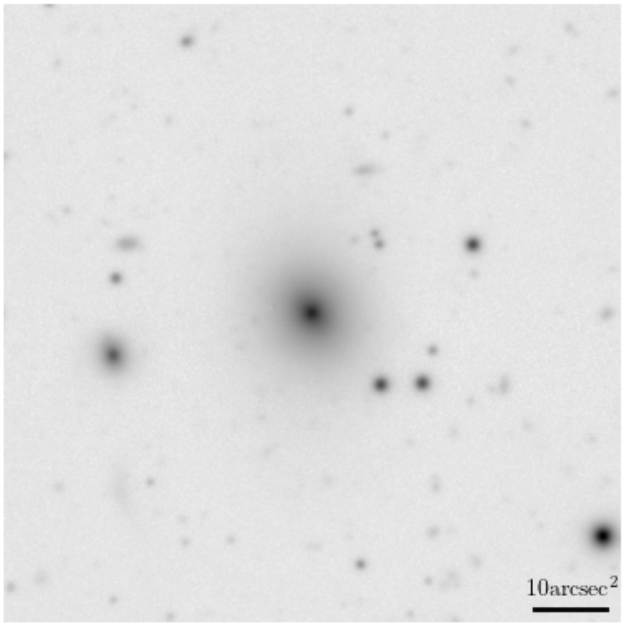


Figure 9. Log scaled image of galaxy (id = 381229) that was "reasonably" misclassified as featureless due to subtle feature.

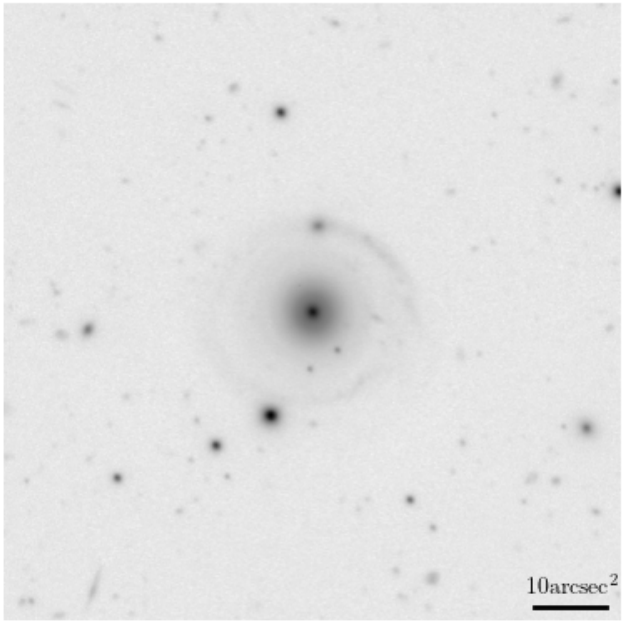


Figure 10. Log scaled image of galaxy (id = 105738) that was "unreasonably" misclassified as featureless with an obvious feature.

Removing these samples from the dataset increases the AUC-ROC to an acceptable 0.75 Mandrekar (2010).

6 DISCUSSION

The time dependent performance of the network provides evidence in support of our hypothesis that networks trained on more effectively processed images would learn faster. This is in agreement with de Raad et al. (2021)

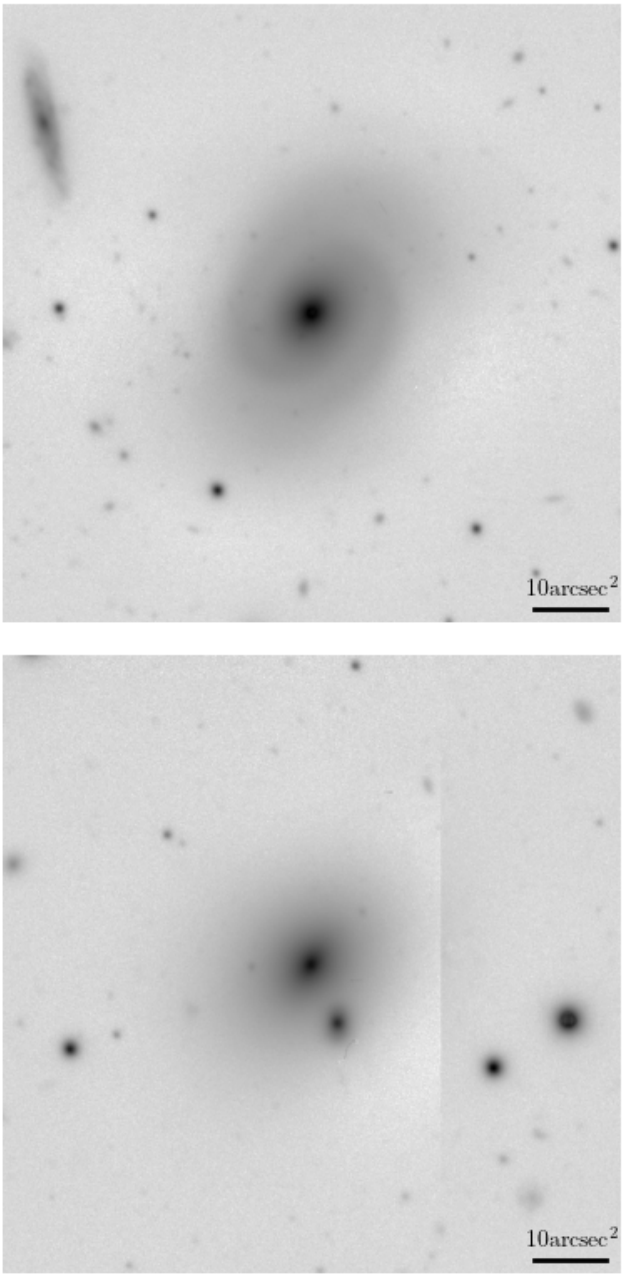


Figure 11. Log scaled images of misclassified galaxies with no features (top) galaxy (id = 107137) with spiral arms (bottom) galaxy (id = 107137) with camera artifact.

We have also provided evidence that networks operating on data that has had information removed will reach a lower maximum accuracy. However, contrary to our hypothesis, we found that on linearly scaled images, with small data samples, the maximum performance is lower than log scaled images, despite them having the same amount of information. This likely resulted from the initial layers of the network being "wasted" doing what is effectively done by image scaling, meaning there are less parameters available for actual feature detec-

tion. It is possible that with a larger training set, or a deeper network, this difference between linear and log scaling would decrease.

The final network did outperform Sánchez et al. (2023) which had a ROC-AUC of 0.64. However, the overall performance of our network was very low, this was likely due to the low true negative rate. Many of the false negatives identified were galaxies that had spiral arms or were contaminated. Removing these images improved the overall performance of the network to an acceptable standard of AUC-ROC = 0.75.

There are some simple improvements that could be made to increase the performance of the network. The HSC-SSP pipeline provides information on image contamination by bright stars and other astronomical objects. Images that have high levels of contamination could be rejected and instead manually classified. This rejection process has been successfully implemented by Kado-Fong et al. (2018). Additionally, many of the image artifacts are likely results of features of the early HSC-SSP pipeline, later data releases are likely to have less of these, and artifact detection is already implemented into the pipe-line (Bosch et al. 2018). Additionally, it is worth noting that the HSC-SSP data is collected for high Redshift science (Aihara et al. 2017), and is not optimized for imaging tidal features. Moving forward, the HSC-SSP "pipeline" could be modified, or data from surveys specifically designed for tidal feature detection could be used.

Another simple improvement to the performance would be to experiment with the architecture to develop an architecture specifically designed for tidal feature detection, as opposed to using AlexNet.

While these above changes would help to improve the network performance, detection of features in spiral galaxies would likely still be difficult. An obvious improvement would be to build a deeper network, this would however require more data and training time. Alternately, a galaxy morphology classifier could be used to detect which galaxies are spiral galaxies and separate them out (such as Walmsley et al. 2019), these could then be processed manually or by a network specifically designed and trained to detect features in spiral galaxies. With enough data, it could even be possible to integrate the processes and build a network that classifies tidal features and galaxy morphology simultaneously as a multi-label classifier.

Due to the challenge of having very few examples of classified galaxies, developing a semi-supervised network that trains on both labelled and unlabelled data could be beneficial (Johnson & Zhang 2015).

ACKNOWLEDGEMENTS

We would like to thank Professor Croom and Mr Rutherford for their continued mentorship throughout the project. Additionally we thank Dr Van de Sande for helping classify the galaxy images.

The Hyper Suprime-Cam (HSC) collaboration includes the astronomical communities of Japan and Taiwan, and Princeton University. The HSC instrumentation and software were developed by the National Astronomical Observatory of Japan (NAOJ), the Kavli Institute for the Physics and Mathematics of the Universe (Kavli IPMU), the University of Tokyo, the High Energy Accelerator Research Organization (KEK), the 19 Academia Sinica Institute for Astronomy and Astrophysics in Taiwan (ASIAA), and Princeton University. Funding was contributed by the FIRST program from Japanese Cabinet Office, the Ministry of Education, Culture, Sports, Science and Technology (MEXT), the Japan Society for the Promotion of Science (JSPS), Japan Science and Technology Agency (JST), the Toray Science Foundation, NAOJ, Kavli IPMU, KEK, ASIAA, and Princeton University.

REFERENCES

Abadi M., et al., 2015, TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems, https://www.tensorflow.org/
Aihara H., et al., 2017, Publications of the Astronomical Society of Japan, 70
Atkinson A. M., Abraham R. G., Ferguson A. M. N., 2013, The Astrophysical Journal, 765, 28
Blanton M., Cen R., Ostriker J. P., Strauss M. A., Tegmark M., 2000, The Astrophysical Journal, 531, 1
Bosch J., et al., 2018, PASJ, 70, S5
Buda M., Maki A., Mazurowski M. A., 2018, Neural Networks, 106, 249
Cheng T.-Y., et al., 2020, Monthly Notices of the Royal Astronomical Society, 493, 4209
Cole S., Aragon-Salamanca A., Frenk C. S., Navarro J. F., Zepf S. E., 1994, Monthly Notices of the Royal Astronomical Society, 271, 781
Cole S., Lacey C. G., Baugh C. M., Frenk C. S., 2000, Monthly Notices of the Royal Astronomical Society, 319, 168
Collaboration L. S., et al., 2009, LSST Science Book, Version 2.0 (arXiv:0912.0201)
Hendel D., Johnston K. V., 2015, Monthly Notices of the Royal Astronomical Society, 454, 2472
Johnson R., Zhang T., 2015, Advances in neural information processing systems, 28
Kado-Fong E., et al., 2018, The Astrophysical Journal, 866, 103
Krizhevsky A., Sutskever I., Hinton G. E., 2017, Communications of the ACM, 60, 84
Mandrekar J. N., 2010, Journal of Thoracic Oncology, 5, 1315
Pop A.-R., Pillepich A., Amorisco N. C., Hernquist L., 2018, Monthly Notices of the Royal Astronomical Society, 480, 1715
Steinmetz M., Navarro J. F., 2002, New Astronomy, 7, 155
Sánchez H. D., 2021, in Debating the potential of machine learning in astronomical surveys. IAP colloquium 2021, Paris
Sánchez H. D., et al., 2023, Monthly Notices of the Royal Astronomical Society, 521, 3861
Tal T., van Dokkum P. G., Nelan J., Bezanson R., 2009, The Astronomical Journal, 138, 1417
Taylor L., Nitschke G., 2018, in 2018 IEEE symposium series on computational intelligence (SSCI). pp 1542–1547
Toomre A., Toomre J., 1972, ApJ, 178, 623
Walmsley M., Ferguson A. M. N., Mann R. G., Lintott C. J., 2018, Monthly Notices of the Royal Astronomical Society, 483, 2968
Walmsley M., et al., 2019, Monthly Notices of the Royal Astronomical Society, 491, 1554
Yoon Y., Lim G., 2020, The Astrophysical Journal, 905, 154
de Raad K., et al., 2021, in 2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI). pp 655–658, doi:10.1109/ISBI48211.2021.9433952